Feedback and Solutions

F78AA Actuarial and Financial Mathematics A
Feedback and Solutions to the December 2018 Exam

In the following we provide solutions and comments (in italics) for the 2018/19 exam. In each case we indicate a potential solution, there may well be alternative ways of addressing the question. These were – if correctly done – also rewarded with marks.

1. *This question was seen on Tutorial Sheet 2, and most students scored (almost) full marks. The question mainly tests students' ability to do numerical calculations.*

   Denote discounting/accumulation from time t to time s by $v(s,t)$.

   (a) The accumulation to $t = 5$ is:

   $$\begin{aligned}&1000v(5;0) - 500v(5;1.5) + 100v(5;2) - 500v(5;2.5) - 500v(5;4)\\ =&1000(1.08)^5 - 500(1.08)^{3.5} + 100(1.08)^3 - 500(1.08)^{2.5} - 500(1.08)^1\\ =&-205.3455\end{aligned}$$

   or -205.35 rounded to two decimals.

   (b) The present value at $t = 0$ is the accumulation to $t = 5$ rolled back to time 0, i.e.

   $$-205.3455 \times v(0;5) = -205.3455 \times 1.08^{-5} = -139.7547.$$

   Students may of course discount all payments and them sum the discounted values.

   (c) The value at $t = 2$ is the accumulation to $t = 5$ rolled back to time 2, i.e.

   $$-205.3455 \times v(2;5) = -205.3455 \times 1.08^{-3} = -163.0099.$$

   The obvious alternative method of solving this question is to accumulate/discount all payments to time $t = 2$. A common mistake was to forget the payments happening after time 2.

2. *This question was a variation on tutorial questions and lectures. Students who had prepared well and could correctly recall how to calculate the TWRR did well in the question.*

   Let i_1, i_2, i_3 be the effective returns in each calendar year.

$$\begin{aligned} 1+i_1 &= (16.9-1.5)/14 = 1.1 \\ 1+i_2 &= (16.893-0.5)/16.9 = 0.97 \quad \text{(i.e. a loss-making year)} \\ 1+i_3 &= (20.920160-2)/16.893 = 1.12 \end{aligned}$$

$$\begin{aligned} (1+TWRR)^3 &= (1+i_1)(1+i_2)(1+i_3) = 1.19504 \\ \Rightarrow TWRR &= 0.061192 \quad \text{or about } 6.12\% \end{aligned}$$

3. *Part (a) of the question tests students' ability to understand and prove simple actuarial formulae. Part (b) of the question addresses the ability to decompose cashflows so that symbols from the international actuarial notation apply.*

   As regards (a), students were on the whole unable to do these calculations correctly. There seem to be multiple reasons for this, chief amongst them being:

   - *Lack of familiarity with the concept of "proof" in mathematics.*
   - *Inability to do even simple manipulations of formulae such as multiplying powers or factoring out common terms.*
   - *Lack of knowledge of standard actuarial notation and formulae.*

   We should not forget to mention that the required calculation appeared in slightly different form in lectures (Thursday of Week 3) and in the printed notes (p. 42).

   As for (b), most students were unable to decompose the cash flows into simple building blocks that can be attacked using standard actuarial notation. The idea of the solution was explained in detail in the Thursday lecture of Week 3 in the form of a worked example. As a final remark: This question appeared as Question 4 (b) in the December 2016 exam for F78AA (available incl. solutions on Vision under "Learning Materials") so that a bit of revision may indeed have helped.

   (a) We have

   $$(I\ddot{a})_{\overline{n}|} = 1 + 2(1+i)^{-1} + 3(1+i)^{-2} + \cdots + n(1+i)^{-(n-1)}.$$

   Therefore

   $$(1+i)^{-1}(I\ddot{a})_{\overline{n}|} = (1+i)^{-1} + 2(1+i)^{-2} + 3(1+i)^{-3} + \cdots + n(1+i)^{-n}.$$

   Taking the difference $(I\ddot{a})_{\overline{n}|} - (1+i)^{-1}(I\ddot{a})_{\overline{n}|}$ yields

   $$\left(1 - \frac{1}{1+i}\right)(I\ddot{a})_{\overline{n}|} = 1 + (1+i)^{-1} + (1+i)^{-2} + \cdots + (1+i)^{-(n-1)} - n(1+i)^{-n}.$$

   Recalling that $v = (1+i)^{-1}, d = 1 - v$ and

   $$\ddot{a}_{\overline{n}|} = 1 + (1+i)^{-1} + (1+i)^{-2} + \cdots + (1+i)^{-(n-1)}$$

   finishes the proof.

(b) The P.V. at the start of the payment stream is

$$\begin{aligned}\text{P.V.} =& 10 + 12v + 14v^2 + \cdots + 48v^{19} \\ =& 10 + (10+2)v + (10+4)v^2 + \cdots + (10+38)v^{19} \\ =& 10\ddot{a}_{\overline{20|}} + 2v\left(1 + v + 2v^2 + 3v^3 + \cdots + 19v^{18}\right) \\ =& 10\ddot{a}_{\overline{20|}} + 2v(I\ddot{a})_{\overline{19|}}.\end{aligned}$$

[Alternatively $(8+2)v^0+(8+4)v^1+\ldots+(8+40)v^{18} = 8\ddot{a}_{\overline{20|}}+2(I\ddot{a})_{\overline{20|}}$.]
Calculating at $i = 0.03$, we find

$$\ddot{a}_{\overline{20|}} = 15.32380, \quad \ddot{a}_{\overline{19|}} = 14.75351$$

and

$$(I\ddot{a})_{\overline{19|}} = \frac{\ddot{a}_{\overline{19|}} - 19v^{19}}{1 - 1/1.03} = \frac{14.75351 - 19v^{19}}{0.029126} = 134.5205.$$

Therefore,

$$\text{P.V.} = 10 \times 15.32380 + 2v \times 134.5205 = 414.443.$$

4. *This question was a variation on some tutorial questions.*

   Part (a) is a standard theoretical result. However, many students achieved the correct answer (100) by calculating the price numerically.

   In (b), well prepared students understood that they are looking forwards 25 from time 3 to time 28 and wrote down the correct equation for P before commencing the numerical calculations, and knew how to check if CGT was payable. Other candidates struggled in a variety of ways.

   (a) The gross redemption yield given (4%) is equal to the coupon rate, and coupons are payable annually in arrears.
   Therefore the bond price is 100.

   (b) The original bond duration is 28 years. 25 years remain at the time of the sale to Kelly.
   Let $I = 0.25$ be the rate of income tax, $C = 0.4$ be the rate of CGT, P be the price paid by Kelly at time 3, $i = 0.04$ is the yield after taxes, and $v = 1/(1+i)$.
   Then $v^{25} = 0.4549960$ and $a^{i}_{\overline{25}|} = 17.03138$.
   The equation of value is

$$\begin{aligned} P &= 4(1-I)a^{i}_{\overline{25}|} + (100 - C(100-P))v^{25} \\ \Rightarrow P(1-Cv^{25}) &= 4(1-I)a^{i}_{\overline{25}|} + 100(1-C)v^{25} \\ \Rightarrow P &= \frac{4 \times 0.75 \times 17.03138 + 100 \times 0.6 \times 0.4549960}{1 - 0.4 \times 0.4549960} \\ &= \frac{78.39389}{0.8180016} \\ &= 95.83586 \end{aligned}$$

   (c) John pays 100 and received back 95.84 at time 3. Therefore he makes a capital loss and no capital gains tax is payable. John's equation of value therefore is (net of tax):

$$\begin{aligned} 100 &= (1-0.2) \times 4 \times a_{\overline{3}|} + 95.83586v^3 \\ \Rightarrow 100 &= 3.2a_{\overline{3}|} + 95.83586v^3 \quad \text{at rate of interest } j \text{ to be determined.} \end{aligned}$$

   So we solve

$$f(j) = 3.2a_{\overline{3}|} + 95.83586v^3 - 100 = 0.$$

   We try:

   - $f(0.018) = 0.1061$
   - $f(0.019) = -0.1792$

   So $f(j)$ changes sign between 0.018 and 0.019. $f(j)$ is continuous in j, so there must be a solution to $f(j) = 0$ between 0.018 and 0.019.

5. *This was a slightly more challenging variation on tutorial questions. Again, there was a wide spread of results in this question with low marks indicating lack of preparation on the part of many candidates.*

   Candidates who performed well typically kept a cool head and worked things out from first principles in line one of their solutions, and then proceeded carefully in small steps. Other candidates tried to guess what the right formula would be in line one of their solutions or tried to jump 2 or 3 steps in one go often coming unstuck in the process.

   (a) i. Payment at time t is $200 \times 1.05^{t-1}$.
   Hence the payment at time 20 is $200 \times 1.05^{19} = 505.39$.

   ii. Let $V(t)$ be the present value at time 0 of the payments up to time t:

$$\begin{aligned} V(t) &= \sum_{s=1}^{t} 200 \times 1.05^{s-1} v^s \\ &= \frac{200}{1.05} a_{\overline{t}|} \quad \text{at rate } j = 1.08/1.05 - 1 = 0.028571 \end{aligned}$$

   Now solve:

$$\begin{aligned} 2000 &= \frac{200}{1.05} a_{\overline{t}|} \text{ as if } t \text{ is continuous} \\ \Rightarrow a_{\overline{t}|} &= \frac{1.05 \times 2000}{200} = 10.5 \text{ at rate } j \\ \Rightarrow 1 - v^t &= 10.5j \quad \text{where } v = 1/(1+j) \\ \Rightarrow v^t &= 1 - 10.5j \\ \Rightarrow t &= \log(1 - 10.5j)/\log v \\ &= 12.66. \end{aligned}$$

   Hence $DPP = 13$ is the first time when the initial loan is fully repaid with some cash left over.

   iii. The surplus at time 13 is

$$\left(200/1.05 \times a^{j}_{\overline{13}|} - 2000\right) \times 1.08^{13} = 120.5831.$$

   The PV at 6% at time 13 of the accumulated surplus up to time 20 will then be

$$\begin{aligned} PV(13) &= 120.5831 + \sum_{s=14}^{20} 200 \times 1.05^{s-1} v^{s-13} \\ &= 120.5831 + 359.1713 \sum_{u=1}^{7} 1.05^u v^u \\ &= 120.5831 + 359.1713 a^{j}_{\overline{7}|} \quad \text{at rate } j = 1.06/1.05 - 1 = 0.009524 \\ &= 120.5831 + 359.1713 \times 6.740775 \\ &= 2541.6760. \end{aligned}$$

Hence the accumulation of the profit up to time 20 is

$$2541.6760 \times 1.06^7 = 3821.74.$$

(b) i.

$$\begin{aligned} AV &= \frac{200}{1.05}\sum_{s=1}^{20} 1.05^s 1.06^{20-s} - 2000 \times 1.08^{20} \\ &= \frac{200}{1.05} a^{j}_{\overline{20}|} \times 1.06^{20} - 9321.914 \quad \text{where } j = 0.009524 \\ &= \frac{200}{1.05} \times 18.13237 \times 3.207135 - 9321.914 \\ &= 1754.840 \end{aligned}$$

ii. This is less than (a)(iii) because the deposit rate being earned by the early cashflows is now at 6%. So the investor in net terms is earning less interest.
In (b) the early cashflows are effectively earning 8% interest up to the DPP.
Alternative explanation: every pound used to pay off the loan early saves an extra 2% interest per annum.

6. *This is a conceptually and mathematically more challenging question as only algebraic manipulations are required and students are not asked to plug numbers into a given formula.*

   Parts (a) and (b) of the question appear as Question 5 on Tutorial Sheet 5, where an extensive solution is provided. Hardly anyone has scored full marks here, though some students appear to have memorised the solution and reproduced it with suitable mistakes.

   The main issue was again a severe lack of basic mathematical skills:

   - *No knowledge of what "equation of value" means.*
   - *Inability to add two functions f_1, and f_2. Students seem to be unaware of the fact that the addition of functions is defined as a pointwise operation.*
   - *As a second-year AMS student one should also know that in general $f(x_1 + x_2) \neq f(x_1) + f(x_2)$.*
   - *Moreover, it is usually a bad idea to use the same notation for two distinct objects: using the same symbol f for two very different equations of value does not make much sense.*
   - *Especially in (b), students seem unable to factorise a cubic equation into a linear and a quadratic term.*
   - *Students should note that in general $(1 + i)^n \neq 1 + ni$ and that the latter can only be used as an approximation. We stress that "approximation" does not mean "equality", a fact that was not only mentioned and explained in lectures (Thursday of Week 5) but was almost the sole subject of the first computer lab.*

   As for (c), Steffensen's Theorem was discussed in lectures (Thursday of Week 5) and can be found in the printed notes on p. 64. For the most part, students appear to be unable to state the theorem correctly, being confused by "necessary" vs. "sufficient" conditions. Note that the statement of the theorem does not claim there is a unique yield if and only if *the cashflows change sign once. Moreover, several students appear not to know what a "unique" solution means.*

   (a) Let $f_1(i) = 0$ and $f_2(i) = 0$ be the two equations of value associated with the first and second series of cash flows, respectively, both at time 0 rather than at time t_n and s_m, respectively. That is:

$$f_1(i) := a_1 v^{t_1} + a_2 v^{t_2} + \cdots + a_n v^{t_n} = \sum_{k=1}^{n} a_k v^{t_k} = 0,$$

$$f_2(i) := b_1 v^{s_1} + b_2 v^{s_2} + \cdots + b_n v^{s_n} = \sum_{l=1}^{m} b_l v^{s_l} = 0.$$

We know that $f_1(i^\star) = f_2(i^\star) = 0$.
If we create a new set of cashflows simply by combining the two sets, the new equation of value at time 0 is

$$f(i) := f_1(i) + f_2(i) = 0.$$

Obviously, $i^\star$ satisfies this equation.

(b) The two equations of value are:

$$\begin{aligned} f_1(i) &:= (1+i) - 1.1 = 0, \quad \text{(accumulation at time 1)} \\ f_2(i) &:= -1.1(1+i) + 1.21 = 0 \quad \text{(accumulation at time 3)} \end{aligned}$$

It is easily checked that $i^\star = 0.1$ is the unique real solution ≥ -1 of each of these. Notice that $f_2(i) = -1.1 f_1(i)$, so when we write down the equation of value of the combined series of cashflows,

$$f(i) = f_1(i) + f_2(i)v^2 = f_1(i)(1 - 1.1v^2) = 0.$$

Then for $f(i) = 0$, either $f_1(i) = 0$ or $1 - 1.1v^2 = 0$.
Solving the latter equation gives a second real solution $j = 1.11^{1/2} - 1 \approx 0.0488088 \geq -1$. Hence, while there is a solution to the combined equation of value, it is not unique. Thus the yield is undefined.

(c) **Steffensen's Theorem:** Suppose that a financial contract has cashflows $c_1, c_2, \ldots, c_n$ at times $t_1 < t_2 < \ldots < t_n$. Then, if the cashflows change sign exactly once, the yield exists.

This covers the important case of buying a financial asset or taking out a loan. There is a single payment up front (negative), in return for a stream of return payments in future (positive).